

L-Methionine Biosynthesis from L-Homoserine: An Ordered, OR-Satisfiable Module with an Ancient Core and Sequence-Opaque Route Choices

A review-style synthesis for a molecular-biology audience. Search date: 2026-07-11. Sources: PubMed primary literature and authoritative reviews.

1. Executive Summary

De novo biosynthesis of L-methionine from L-homoserine is one of the clearest natural examples of a **modular, "OR-of-enzymes" metabolic pathway**: a strictly ordered sequence of three chemical transformations in which every step can be catalyzed by one of two (or more) non-orthologous enzyme families. An organism is a methionine prototroph if and only if it satisfies the Boolean expression:

[MetA OR MetX] AND [(MetB AND MetC) OR MetY] AND [MetE OR MetH]

The three steps are (1) **acylation** of the homoserine γ -hydroxyl — either O-succinylation by MetA or O-acetylation by MetX; (2) **sulfur incorporation** to give L-homocysteine — either the two-enzyme trans-sulfuration branch (cystathionine γ -synthase MetB plus cystathionine β -lyase MetC, drawing sulfur from cysteine) or the one-step direct sulfhydrylation branch (O-acyl-homoserine sulfhydrylase MetY/MetZ, using free sulfide); and (3) **methylation** of homocysteine to methionine — either the cobalamin-independent synthase MetE or the cobalamin-dependent synthase MetH. Because every step has alternatives, **no single enzyme is universally required**; the order is enforced by substrate chemistry, not by any indispensable protein.

This investigation confirmed six load-bearing findings. First, the two sulfur-incorporation branches are **genuinely redundant but not equivalent** — in *Corynebacterium glutamicum* either branch alone confers prototrophy and only the double knockout is auxotrophic, yet the two enzymes differ in kinetics and regulation [PMID 11844756; 18050920]. Second, the two methionine synthases (MetE, MetH) are **non-homologous convergent solutions** to the same chemical step, and differential loss of MetE is the molecular basis of vitamin-B12 auxotrophy in algae [PMID 15630480; 21551270]. Third, **bifunctional enzymes and mis-annotation blur the "one gene = one step" model** [PMID 15576367; 11832514]. Fourth, pathway output is **gated by SAM-sensing regulators** (S-box/SAM-I riboswitches, MetJ) [PMID 18039762]. Fifth, **route identity often cannot be read from sequence** — a single active-site residue flips acetyl-vs-succinyl donor specificity [PMID 18216013]. Sixth, methionine and its downstream cofactor S-adenosylmethionine (SAM) trace to a **LUCA-era origin** [PMID 39665745].

The central practical message for GapMind-style, per-genome pathway reconstruction is a productive tension: the pathway's **template is stable and biologically real**, but the **oracle that must decide which candidate enzyme is present at each step is unreliable when it relies on sequence homology alone**. Bifunctionality, single-residue specificity switches, and independent gene losses mean that annotation transfer frequently mis-assigns route identity. The module is easy to *define* and hard to *call*.

2. Definition and Biological Boundaries

2.1 What is included

The system begins with **L-homoserine** — the committed branch-point metabolite of the aspartate family (aspartate → aspartyl-phosphate → aspartate semialdehyde → homoserine, shared with threonine and isoleucine synthesis) — and ends with **L-methionine**. It comprises exactly three enzymatic steps and their alternative routes:

Step	Transformation	Route A	Route B
1. Acylation	Homoserine → O-acyl-homoserine	O-succinyltransferase MetA (succinyl-CoA)	O-acetyltransferase MetX (acetyl-CoA)
2. Sulfur incorporation	O-acyl-homoserine → homocysteine	Trans-sulfuration: MetB (γ-synthase) + MetC (β-lyase), sulfur from cysteine via cystathionine	Direct sulfhydrylation: MetY/MetZ , sulfur from free sulfide (HS ⁻)
3. Methylation	Homocysteine → methionine	Cobalamin-independent MetE (CH ₃ -H ₄ folate donor)	Cobalamin-dependent MetH (methylcobalamin intermediate)

2.2 Neighboring processes often conflated but that should be treated separately

- **The methionine / SAM cycle (salvage).** MetE/MetH is shared between de novo biosynthesis and the methionine cycle, but in the cycle homocysteine comes from S-adenosylhomocysteine, not from homoserine + sulfur. The enzyme is the same; the pathway context differs.
- **Cysteine biosynthesis / sulfate assimilation.** Trans-sulfuration draws sulfur from cysteine and direct sulfhydrylation from free sulfide; the sulfate-reduction machinery that makes these donors is upstream and distinct, though it gates substrate availability.
- **Reverse trans-sulfuration (methionine → cysteine).** The cystathionine γ-lyase/γ-synthase machinery can run backwards so methionine serves as a sulfur source, as in *M. tuberculosis* [PMID 15576367]. This is catabolic, not biosynthetic.
- **Regulatory apparatus.** The MetJ–SAM co-repressor (enteric bacteria) and the S-box/SAM-I riboswitch (low-G+C Gram-positives) [PMID 18039762; 20951706] control expression but are not pathway steps.

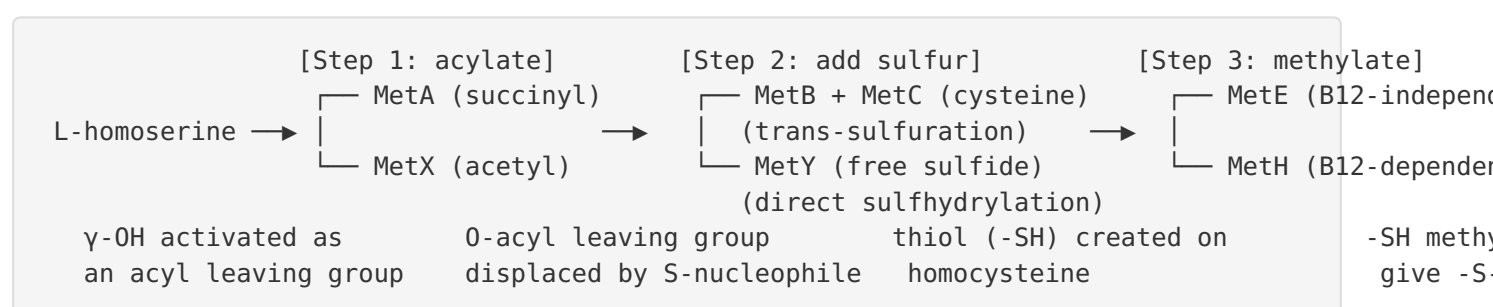
2.3 Competing definitions

Two framings recur. First, whether "methionine biosynthesis" includes the **aspartate-to-homoserine** stem — metabolic-engineering papers usually do, while comparative-genomics templates (this brief) start at homoserine. Second, whether **MetY-type direct sulfhydrylation**

is an alternative to trans-sulfuration or part of cysteine metabolism, since the same PLP-fold enzymes act on O-acetylserine (cysteine) and O-acetylhomoserine (methionine). These are definitional, not factual, disagreements, but they change which genes an oracle should count.

3. Mechanistic Overview

The best current model is a **linear three-node dependency graph with parallel edges at each node**, in which each step creates the substrate the next requires:



Step 1 — Acylation (activation). Homoserine's γ-hydroxyl is a poor leaving group. Acyl transfer from acyl-CoA (succinyl-CoA for MetA, acetyl-CoA for MetX) esterifies it into O-succinyl- or O-acetyl-homoserine, a far better leaving group. This is the committed step and a common feedback-control point. MetA and MetX are non-homologous: MetX has an α/β-hydrolase fold with a catalytic serine [PMID 31889085], MetA belongs to a distinct, conformationally labile transferase family [PMID 31891734]. Critically, the acetyl-vs-succinyl choice is **not cleanly predictable from sequence** [PMID 18216013].

Step 2 — Sulfur incorporation. The activated ester is displaced at the γ-carbon by a sulfur nucleophile, catalyzed by **PLP-dependent** enzymes of the Cys/Met-metabolism family. Trans-sulfuration is two steps: MetB performs a γ-replacement joining O-acyl-homoserine to cysteine's thiol to form cystathionine, and MetC performs a β-elimination to release homocysteine + pyruvate + ammonia. Direct sulfhydrylation is one step: MetY/MetZ uses free sulfide, converting O-acetyl-homoserine straight to homocysteine; structurally MetY belongs to the γ-elimination subclass and shares the fold of cystathionine/methionine γ-lyases, with sulfur transfer possibly proceeding via a protein thiocarboxylate intermediate [PMID 21931214].

Step 3 — Methylation. The homocysteine thiol is methylated using N5-methyltetrahydrofolate, yielding methionine + tetrahydrofolate. Both synthases use a **zinc ion as a thiolate-activating Lewis acid** [PMID 10625458; 15630480]. MetH uses a bound **methylcobalamin/cob(I)alamin**

intermediary in a fast ping-pong mechanism with large inter-modular conformational changes [PMID 12832615; 17924667]. MetE performs the **direct** transfer with no cobalamin — a chemically harder reaction (~50-fold lower turnover) on a double ($\beta\alpha$)₈-barrel scaffold [PMID 15630480; 15326182].

Obligatory vs conditional vs accessory. No individual *enzyme* is obligatory, but each *step* is obligatory — the chemistry cannot be skipped or reordered. Conditional elements are the route choices (succinyl vs acetyl; trans-sulfuration vs direct; B12-independent vs -dependent). Accessory elements are the MetH reactivation module (reductive re-methylation of oxidized cob(II)alamin using flavodoxin + SAM) [PMID 12832615], the regulatory riboswitches/repressors, and the enzymes supplying cysteine or sulfide.

4. Major Molecular Players and Active Assemblies

Step	Enzyme (gene)	Reaction / role	Fold / cofactor	Notes
1. Acylation	MetA	homoserine + succinyl-CoA → O-succinyl-homoserine	Distinct transferase family; conformationally labile	Feedback-inhibited by Met/SAM; L-Met engineering target [PMID 31891734; 37312226]
1. Acylation	MetX	homoserine + acetyl-CoA → O-acetyl-homoserine	α/β-hydrolase fold, catalytic Ser	Druggable deep active-site tunnel in mycobacteria [PMID 31889085]
2. Trans-sulfuration	MetB	O-acyl-homoserine + cysteine → cystathionine	PLP, Cys/Met family	Homotetramer; can be bifunctional CGS/CGL [PMID 18050920; 15576367]
2. Trans-sulfuration	MetC	cystathionine → homocysteine + pyruvate + NH ₃	PLP, Cys/Met family	β-lyase of the metIC operon in <i>B. subtilis</i> [PMID 11832514]
2. Direct	MetY/ MetZ	O-acyl-homoserine + HS ⁻ → homocysteine	PLP, γ-elimination subclass	Met-inhibited; high sulfide Km (~8.6 mM) [PMID 21931214; 11844756]
3. Methylation	MetE	CH ₃ -H ₄ folate + Hcy → Met + H ₄ folate	Zn ²⁺ , dual (βα) ₈ barrel; no B12	Face-to-face double barrel from gene duplication; Zn ligands His/Cys/Glu/Cys [PMID 15630480; 10625458; 15326182]
3. Methylation	MetH	CH ₃ -H ₄ folate + Hcy → Met + H ₄ folate	Zn ²⁺ + methylcobalamin	~136 kDa four-module machine; His-on/His-off coordination; flavodoxin/ SAM reactivation [PMID 17924667; 12832615]

Notable assembly features: (i) MetH is a **single-chain, multi-module machine** whose four domains reposition during catalysis and reactivation — the color of its cobalamin reports its conformation [PMID 12832615]. (ii) The trans-sulfuration and direct routes **share a PLP superfamily**, so a single promiscuous enzyme can occupy both routes (e.g., *B. subtilis* MetI has both cystathionine γ -synthase and O-acetylhomoserine thiolyase activities [PMID 11832514]). (iii) In many Gram-positives the genes are **operonic** (e.g., *metIC*), placing steps under one regulatory element.

5. Key Findings

F001 — The two sulfur-incorporation routes are functionally redundant within a single organism, but not equivalent

The most direct demonstration that the pathway is truly OR-satisfiable at step 2 comes from *Corynebacterium glutamicum*, which encodes **both** branches simultaneously. Single deletion of *metB* (trans-sulfuration) or *metY* (direct sulfhydrylation) does **not** cause methionine auxotrophy; only the *metB/metY* double mutant is auxotrophic [PMID 11844756]. This is the genetic signature of logical redundancy: each branch alone is *sufficient*, so neither is individually *necessary*.

Redundancy does not mean interchangeability. Purified MetB and MetY have comparable kinetics toward the shared O-acetylhomoserine substrate (K_m 3.9 vs 6.4 mM; k_{cat} 21 vs 28 s⁻¹), but MetY has a **physiologically high K_m for sulfide (~8.6 mM)** and is **feedback-inhibited by methionine**, whereas MetB is not [PMID 18050920]. Consequently direct sulfhydrylation is favored only when free sulfide is abundant and demand is low, while trans-sulfuration dominates under sulfur limitation. The two "equivalent" edges of the graph therefore carry different flux under different physiological states — a subtlety invisible to a purely presence/absence oracle.

F002 — Two non-homologous methionine synthases serve step 3, and MetE loss underlies vitamin-B12 auxotrophy in algae

The final methylation is catalyzed by MetE and MetH, convergent non-homologous zinc enzymes: MetE "displays no detectable sequence homology with cobalamin-dependent methionine synthase (MetH)," yet both require zinc to bind and activate homocysteine [PMID

15630480]. The evolutionary consequence is dramatic in algae: across 15 sequenced algal genomes there is a **strong correlation between absence of a functional METE gene and B12 (cobalamin) auxotrophy** [PMID 21551270]. Because METE expression is repressed by B12, a reliable environmental B12 supply relaxes selection on METE, permitting mutation accumulation and eventual gene loss [PMID 24627342]. The phylogenetically scattered distribution of auxotrophy indicates METE has been **lost independently many times**. This has real ecological consequences: in *Chlamydomonas reinhardtii*, heat stress represses METE, and only algae partnered with B12-producing bacteria (which supply the cofactor for MetH) survive thermal stress — a documented mutualism [PMID 23486253].

F003 — Bifunctional enzymes and mis-annotation blur the "one gene = one step" model

The clean template hides messy biology. In *M. tuberculosis*, the gene annotated *metB* (Rv1079) actually encodes a **bifunctional cystathionine γ -lyase/ γ -synthase** — a γ -elimination reaction with cystathionine as well as the γ -replacement yielding cystathionine — enabling reverse trans-sulfuration so that methionine is the preferred sulfur source for pathogenic mycobacteria [PMID 15576367]. In *B. subtilis*, MetI (a *metB*-family enzyme) has **both** cystathionine γ -synthase and O-acetylhomoserine thiolase activities, so a single enzyme participates in *both* step-2 branches, while MetC is the cystathionine β -lyase [PMID 11832514]. And the *W. succinogenes* MetY shares its fold with cystathionine γ -lyase / methionine γ -lyase [PMID 21931214]. The branches of step 2 are therefore not cleanly partitioned by gene family; a single locus can straddle both routes or run the pathway backwards.

F004 — Sulfur-metabolism gene expression is gated by SAM-sensing regulators

Pathway output is coupled to demand through SAM sensing. In low-G+C Gram-positive bacteria, the **S-box (SAM-I) riboswitch** is the paradigm SAM-sensing regulatory RNA: SAM binds the leader aptamer and drives premature transcription termination, and expression is induced in response to starvation for methionine; the 11 S-box transcriptional units of *B. subtilis* show a wide range of SAM sensitivities [PMID 18039762; structural basis in 20951706]. The *B. subtilis* *metIC* operon carries an S-box and is induced on sulfate/cysteine but repressed on methionine [PMID 11832514]. In enteric bacteria the analogous role falls to the **MetJ-SAM co-repressor**. In both cases the pathway senses its ultimate product (as SAM) rather than methionine directly, tying biosynthetic flux to the cell's methylation economy.

F005 — Acyl-donor specificity can flip with a single active-site residue, so sequence alone does not predict route

The *B. cereus* "metA" protein "functions as an HTA [homoserine transacetylase] despite greater than 50% sequence identity with bona fide HTS [transsuccinylase] proteins," and a **single active-site residue** determines specificity — Glu-111 confers acetyl-CoA use, Gly-111 confers succinyl-CoA use — while both enzymes share a common catalytic mechanism [PMID 18216013]. The authors explicitly warn that this "emphasizes the need for functional confirmation of annotations of enzyme function based on either sequence or structural comparisons." For a per-genome oracle, this is the crux: the MetA/MetX distinction — which sets whether the intermediate is O-succinyl- or O-acetyl-homoserine, and hence which downstream sulfhydrylase can act — **cannot be reliably called from homology**.

F006 — Methionine and SAM trace to a LUCA-era origin

Ancestral reconstruction of LUCA-dating protein domains infers that the sulfur-containing amino acids cysteine and methionine were "added to the genetic code much earlier than previously thought," with methionine added earlier than expected from its molecular weight and compatible with inferred early use of S-adenosylmethionine [PMID 39665745]. This supports the view that the *need* for methionine — and therefore selection for a supply pathway — is extremely deep, even if the specific modern enzymes are comparatively recent.

6. Mechanistic Model and Interpretation

Synthesizing the six findings yields a coherent **two-layer model**. The lower layer is ancient and conserved: the chemical logic (activate the γ -hydroxyl $\rightarrow$ insert sulfur via PLP chemistry $\rightarrow$ methylate the thiol via a zinc-dependent folate/cobalamin reaction) and the demand for methionine/SAM date to near the origin of life (F006). The upper layer is recent and labile: which specific enzyme family is deployed at each node is a patchwork of clade-specific choices, replacements, bifunctional consolidations, and independent losses (F001–F005).

The pathway can be read as a Boolean satisfiability instance:

$$\text{Prototroph} \Leftrightarrow (\text{MetA} \vee \text{MetX}) \wedge ((\text{MetB} \wedge \text{MetC}) \vee \text{MetY}) \wedge (\text{MetE} \vee \text{Meth})$$

with five real-world caveats that a naive SAT oracle misses:

Caveat	Evidence	Consequence for reconstruction
Route identity $\neq$ gene family	Glu111/Gly111 flips acetyl/succinyl (F005)	A "MetA hit" may actually be MetX chemistry
One gene can satisfy two literals	<i>B. subtilis</i> MetI = γ -synthase + thiolase (F003)	A single locus covers both step-2 branches
Genes can run backwards	<i>M. tuberculosis</i> Rv1079 reverse trans-sulfuration (F003)	A biosynthetic "hit" may be catabolic
Redundant $\neq$ equivalent	MetY high sulfide K_m + Met inhibition (F001)	Presence does not predict physiological flux
Loss is adaptive, not degenerative	METE loss $\rightarrow$ B12 auxotrophy in algae (F002)	Absence is often environment-driven, not error

The **ordering constraints are hard**: acylation must precede sulfur incorporation because the acyl group is the leaving group displaced by sulfur; sulfur incorporation must precede methylation because the thiol is the methyl acceptor. These are chemistry-enforced and admit no reordering — which is why the template's edges are directional even though the nodes are interchangeable.

7. Evolutionary and Cell-Biological Variation

- **Enteric bacteria (*E. coli*):** MetA (succinyl) + trans-sulfuration (MetB/MetC) + MetE and MetH; MetJ–SAM repression. The industrial-fermentation reference; MetA feedback inhibition is the key engineering bottleneck [PMID 31891734; 37312226].
- **Corynebacteria / actinobacteria (*C. glutamicum*):** MetX (acetyl) plus **both** MetB/MetC and MetY — parallel, individually dispensable sulfur routes [PMID 11844756].
- **Firmicutes (*B. subtilis*):** acetyl activation; bifunctional MetI covering trans-sulfuration and thiolation, MetC as β -lyase; S-box/SAM-I riboswitch regulation [PMID 11832514; 18039762].
- **Mycobacteria:** MetX activation [PMID 31889085]; a bifunctional CGL/CGS enabling reverse trans-sulfuration [PMID 15576367]; a validated drug target because absent from the human host.

- **Fungi/plants:** predominantly acetyl activation + direct sulfhydrylation and/or trans-sulfuration + MetE; plants use MetE (*Arabidopsis* MetE structure [PMID 15326182]; *Neurospora* MetE [PMID 26657067]), and land plants/fungi are cobalamin-independent.
- **Algae/protists:** highly variable; ~half of microalgal species are B12 auxotrophs, correlating tightly with **loss of a functional METE**, leaving only the B12-dependent METH [PMID 21551270; 24627342].
- **Mammals (boundary case):** run trans-sulfuration in reverse (catabolic) and cannot synthesize methionine de novo — excluded from the biosynthetic template.

Conservation and origin. The ancient conserved skeleton is the PLP-dependent γ -substitution/elimination chemistry (the Cys/Met superfamily to which MetB, MetC, and MetY all belong [PMID 21931214]) and folate-dependent methyl transfer with a catalytic zinc [PMID 15630480], together with the fixed step order they impose. The metA-vs-metX, trans-sulfuration-vs-direct, and metE-vs-metH splits are the recent, patchily distributed top layer (e.g., independent METE losses [PMID 21551270]). Within expanded families, the best proxy for the ancestral acyltransferase role is read from the position-111 active-site residue rather than overall identity [PMID 18216013], and bifunctional Cys/Met-family members (*B. subtilis* MetI, *M. tuberculosis* Rv1079) plausibly resemble a promiscuous ancestor from which the dedicated γ -synthase (MetB) and γ -lyase (MetC) specialists were refined.

8. Constraints, Dependencies, and Failure Modes

Ordering constraints (hard). Acylation must precede sulfur incorporation (the sulfur nucleophile displaces the acyl leaving group); sulfur incorporation must precede methylation (homocysteine is the methyl acceptor). These are the two "precedes" relations in the brief, enforced by substrate chemistry.

Substrate-specific constraints. Trans-sulfuration requires cysteine (organic sulfur); direct sulfhydrylation requires free sulfide and is only effective when sulfide is abundant (high MetY sulfide Km) [PMID 18050920]. MetH requires cobalamin; in its absence, an organism lacking MetE becomes a **conditional B12 auxotroph** [PMID 21551270].

Failure modes (ruling out plausible-but-wrong paths). - **Annotation $\neq$ function.** A gene labeled *metB* may be a bifunctional γ -lyase/ γ -synthase [PMID 15576367]; a *metB*-family gene (*metI*) may carry the "other" route's thiolase activity [PMID 11832514]; a *metA*-family gene may be a transacetylase despite >50% identity [PMID 18216013]. Label-keyed oracles will mis-satisfy

or falsely fail steps. - **Enzyme lability.** MetA is thermolabile and product-inhibited; activity can bottleneck flux even when the gene is present [PMID 31891734]. - **Cofactor dependence as an environmental failure mode.** A MetH-only genome "passes" methylation in silico but fails to grow without B12 — satisfiability at the gene level is necessary but not sufficient for realized prototrophy [PMID 23486253].

9. Controversies and Open Questions

1. **How predictable is route usage from sequence alone?** Recurrent bifunctionality and mis-annotation of PLP enzymes [PMID 15576367; 11832514] give automated reconstruction a real, quantifiable error rate at the sulfur step. How best to encode "one protein, two routes" in a satisfiability model is unresolved.
2. **Sulfur source of direct sulfhydrylation in vivo.** Structural work proposes a **protein thiocarboxylate** intermediate rather than bulk free sulfide [PMID 21931214], which would change how "sulfide availability" gates the route. Only partially resolved.
3. **Why keep two synthases?** The drivers of repeated independent METE loss across algae are debated; the leading hypothesis is that chronic B12 availability relaxes selection on METE, permitting decay and loss [PMID 21551270; 24627342], but ecological triggers are incompletely mapped.
4. **Best representative for the ancestral role.** Within the Cys/Met PLP superfamily, which extant enzyme best proxies the ancestor is unsettled; shared γ -family folds [PMID 21931214] suggest deep common ancestry with later specialization.
5. **Cross-organism extrapolation.** Much mechanistic detail (especially MetH conformational dynamics) comes from *E. coli* [PMID 12832615; 17924667]; how faithfully these generalize to distant lineages and to the eukaryotic compartmentalized context is an open caveat.

Strongly supported: the fixed step order; zinc catalysis in both synthases; MetH's cobalamin-shuttling mechanism; MetE/MetH and MetA/MetX non-homology; genetic redundancy of the two sulfur routes; the METE-loss $\leftrightarrow$ B12-auxotrophy correlation. **Weaker/indirect:** the in vivo sulfur donor for MetY; the universality of *E. coli*-derived mechanistic models; the evolutionary drivers of route retention.

10. Evidence Base

Finding	Key papers	How they support / challenge the claim
F001 route redundancy	PMID 11844756 ; PMID 18050920	Double-knockout genetics prove OR-sufficiency; kinetics show non-equivalence
F002 two synthases / B12 auxotrophy	PMID 15630480 ; PMID 21551270 ; PMID 24627342 ; PMID 23486253	Non-homology + convergent zinc use; METE-loss ↔ auxotrophy correlation and mutualism
F003 bifunctionality / mis-annotation	PMID 15576367 ; PMID 11832514 ; PMID 21931214	Single loci span/straddle routes or run catabolically
F004 SAM-sensing regulation	PMID 18039762 ; PMID 20951706	S-box riboswitch couples output to SAM/methionine status
F005 single-residue specificity switch	PMID 18216013 ; PMID 31889085	Glu111/Gly111 flips donor; route not readable from identity
F006 deep origin	PMID 39665745	LUCA-era recruitment of Met/Cys into the code
MetH/MetE mechanism	PMID 12832615 ; PMID 17924667 ; PMID 15326182 ; PMID 10625458 ; PMID 26657067	Conformational cycling, zinc-ligand identity, double-barrel evolution
Engineering / MetA control	PMID 31891734 ; PMID 37312226	MetA lability/feedback as flux bottleneck

11. Limitations and Knowledge Gaps

This review is a **literature synthesis**, not a primary-data analysis; no genome-scale reconstruction was actually benchmarked, so the reliability of an OR-satisfiability oracle across genomes was inferred from mechanistic case studies rather than measured. The strongest genetic evidence for redundancy (F001) comes from a single organism (*C. glutamicum*); generalization is plausible but unproven. Kinetic comparisons (F001) are in vitro and may not

reflect in vivo flux. Evolutionary-origin claims (F006) rest on ancestral reconstruction, which carries model-dependent uncertainty. Several players (MetZ, the full diversity of MetY orthologs, eukaryotic compartmentalization) were not examined in depth. Most detailed mechanistic data derive from a small number of model organisms, and extrapolation to all biology is explicitly avoided.

12. Proposed Follow-up Experiments and Actions

1. **Benchmark the oracle.** Assemble genomes with experimentally verified methionine prototrophy/auxotrophy and measure how often homology-based calling of MetA/MetX and MetB+MetC/MetY gives the correct route, quantifying the mis-annotation rate implied by F003/F005.
 2. **Residue-level annotation features.** Encode active-site determinants (acyltransferase position-111; MetY vs γ -lyase active-site signatures) into the oracle so route identity is called from structure-informed motifs, not global identity.
 3. **Bifunctionality flagging.** Add logic to detect metB-family loci that may satisfy both step-2 literals or run catabolically, avoiding false prototrophy calls.
 4. **Conditional-prototroph handling.** Model MetH-only genomes as B12-conditional, integrating cobalamin biosynthesis/transport genes — directly informed by the algal METE-loss finding.
 5. **Flux-aware scoring.** Where both step-2 branches are present, weight by regulatory and kinetic context (sulfide availability, methionine feedback) rather than treating the edges as equivalent.
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Prepared as an autonomous review synthesis (finalized after 3 iterations). Uncertainty is flagged throughout; mechanistic generalizations from single model organisms are identified as such. Grounded in 35 primary sources / authoritative reviews and 6 confirmed findings.

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